

Executive Team Responsibility Matrix

This document details the official division of engineering responsibilities, architecture design, and technical deliverables across our four-student team for the NaviOps platform.

TEAM MEMBER	OFFICIAL ROLE	PRIMARY OWNERSHIP FOCUS	CORE TECH STACK
Smit Sureja	Team Lead & Solutions Architect	Main New Features & Overall System Design GreenPort ESG Ledger, What-If Digital Twin Sandbox Architecture, Sentinel Design	FastAPI, OR-Tools, Next.js, System Architecture
Krish	Frontend & AI Systems Lead	Full Frontend & AI Copilot System All 14 Routes, What-If Studio UI, Gantt Chart, Groq LPU LLM, Tool Calling	Next.js 14, Tailwind CSS, Groq LPU, Agentic Tools
Harshit	Backend & Operations Research Lead	Mathematical Optimization & CP-SAT Engine BAP/QCAP Solver, Simulation Solver Engine, OR-Tools Math	Google OR-Tools, Python 3.11, CP-SAT, Discrete Math
Sarthak	Backend, Telemetry & Security Lead	Core REST Services, Telemetry & Testing FastAPI Repositories, Congestion Index, Sentinel Backend, RBAC, 159 Tests	FastAPI, Pydantic v2, Pytest, JWT/PBKDF2

1. Smit Sureja — Team Lead & Solutions Architect

Smit Sureja — Main New Features & Overall System Design Lead

ARCHITECTURE & MAIN FEATURES

Primary Focus: Overall System Architecture, Domain Modeling, and Flagship "Must-Have" Innovations

- Full-Stack Architecture & Micro-Modular System Design:** Designed the complete architectural blueprint integrating Next.js 14, FastAPI REST services, Google OR-Tools CP-SAT, and Groq LPU with zero cross-service schema corruption.
- GreenPort ESG & Demurrage Financial Ledger (Main Innovation):** Formulated the economic equation converting raw CP-SAT wait-time reductions into hard corporate ledger metrics:

Demurrage Benchmark: \$1,250 / vessel-waiting-hour saved | **Environmental Benchmark:** 0.35 MT CO₂ bunker fuel emissions abated / hour.
- "What-If" Port Digital Twin Sandbox (Core Design & Formulation):** Designed the counterfactual simulation framework allowing terminal operators to stress-test crane outages and berth closures in isolated memory without mutating live operational port schedules.
- Proactive Disruption Sentinel Architecture:** Architected the real-time quayside monitoring model that flags vessels at risk of compounding delays and prescribes 1-click mitigation runbooks.
- Production Hardening & Offline Guarantee:** Engineered pre-seeded datasets, fail-safe offline fallbacks, and deployment configurations ensuring seamless evaluation without external paid APIs.

System Architecture Domain Modeling ESG Financial Ledger Digital Twin Sandbox Offline Orchestration

2. Krish — Frontend & AI Systems Lead

Krish — Frontend Lead & AI Systems Specialist

UI PART & AI PART

Primary Focus: Complete Frontend Implementation, Interactive Dashboards, and Bob Copilot AI Engine

- **Complete Next.js 14 Application & Route Architecture:** Implemented and deployed all 14 application routes with full TypeScript type safety, responsive layout, and enterprise dark-mode UI.
- **Bob Copilot AI Engine & Groq LPU Integration:** Integrated Groq's high-speed inference running openai/gpt-oss-120b, authoring specialized maritime terminal prompts with grounding guardrails.
- **Autonomous Agentic Tool-Calling Framework:** Implemented the AI tool execution dispatcher supporting 10 operational tools including get_vessels, get_congestion, log_disruption, and the new simulate_scenario tool.
- **"What-If" Scenario Studio Interface (/optimization):** Built the interactive tabbed sandbox interface featuring 1-click incident presets (*Crane CR-02 Outage*, *Berth B-01 Maintenance*, *Severe Fog*), manual disruption toggles, live simulation execution, and a 4-card counterfactual delta comparison grid.
- **72-Hour Quayside Gantt Timeline & Executive UI:** Built the quayside berth timeline rendering vessel allocation windows across 4 berths and 6 STS cranes, plus the homepage Sentinel alert banner and Copilot chat UI.

Next.js 14 Tailwind CSS Groq LPU (LLM) Agentic Tool Calling Interactive Gantt TypeScript

3. Harshit — Backend & Operations Research Lead

Harshit — Mathematical Optimization & CP-SAT Engine Lead

CP-SAT & OPTIMIZATION

Primary Focus: Google OR-Tools CP-SAT 72-Hour Optimization Engine & Simulation Solver

- **Google OR-Tools CP-SAT 72-Hour Optimization Engine:** Formulated the exact mathematical model for the joint Berth Allocation Problem (BAP) and Quay Crane Assignment Problem (QCAP), incorporating non-overlapping spatial intervals, crane capacity, and draft constraints.
- **In-Memory Counterfactual Simulation Solver (/api/optimization/simulate):** Programmed the side-effect-free simulation engine that accepts disruption overrides, executes CP-SAT in memory, and computes exact comparative deltas against the baseline schedule.
- **Mathematical Constraint Formulation:** Programmed discrete interval variables, resource cumulative constraints for STS cranes, vessel arrival windows, and berth depth draft safety factors.
- **Solver Delta Computation Engine:** Implemented the mathematical engine calculating changes in wait times, berth utilization, demurrage exposure, and carbon footprint deltas.
- **Optimization Endpoint Architecture:** Designed high-throughput optimization execution endpoints with side-effect-free in-memory safety.

Google OR-Tools CP-SAT Solver Operations Research Discrete Math Python 3.11

4. Sarthak — Backend Engineering, Telemetry & Security Lead

Sarthak — Core Services, Telemetry Scanner & Quality Lead

BACKEND, TELEMETRY & RBAC

Primary Focus: FastAPI Microservices, Congestion Scoring, Disruption Sentinel Telemetry, RBAC & 159-Test Suite

- Core REST Microservices & Domain Repositories: Implemented FastAPI endpoints and repositories for Vessels, Berths, Cranes, Yards, and Disruptions with Pydantic v2 validation.
- Congestion Index Mathematical Calculation Engine: Developed the 0–100 congestion algorithm weighting queue length, crane utilization, and turn-around delay ratios.
- Proactive Disruption Sentinel Engine (/api/disruptions/sentinel): Implemented the quayside telemetry scanner identifying delayed vessels, aggregating financial exposure (\$1,250/hr), and formulating runbook recommendations.
- Enterprise RBAC & Authentication: Implemented PBKDF2 password hashing, JWT bearer tokens, and endpoint permissions across Admin, Operator, Port Master, and Analyst tiers.
- 159-Test Automated Verification Suite: Built and maintained the complete pytest suite (**159/159 passing**) covering API endpoints, data persistence, RBAC permissions, and congestion algorithms.

FastAPIPydantic v2Sentinel ScannerJWT / PBKDF2Pytest (159 Tests)Congestion Formula

Hackathon Presentation Defense: "Who Answers What?"

JUDGE TOPIC / QUESTION	PRIMARY SPEAKER	WINNING KEY TALKING POINTS
"What makes NaviOps unique compared to traditional port software?"	Smit Sureja	Explain the transition from reactive tracking to a proactive Digital Twin with our GreenPort ESG ledger (\$1,250/hr demurrage saved) and What-If Sandbox .
"Walk us through the UI and how the AI Copilot works."	Krish	Demo the Gantt timeline , the What-If Studio presets , and explain the Groq LPU tool-calling agent executing live terminal actions without hallucination.
"How does your CP-SAT solver handle complex scheduling constraints?"	Harshit	Explain that Google OR-Tools CP-SAT solves BAP and QCAP simultaneously as an exact constraint program in ~1.2s without heuristic drift.
"How do you score port congestion and secure the system?"	Sarthak	Detail the 0-100 Congestion Index formula , the Sentinel telemetry risk scanner , and the PBKDF2/JWT enterprise RBAC architecture.

System Verification Sign-Off: 159/159 Pytest passing • 14/14 Next.js routes compiled • 100% Offline ready

PRODUCTION VERIFIED